

$$f(x,y) = \begin{cases} \frac{xy(x^2-y^2)}{x^2+y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$$

(i) Continuous on $\mathbb{R}^2$

(ii) Check for differentiability (Differentiable everywhere)

(iii) Check if $f \in C^1(\mathbb{R}^2)$

(iv) Find if it is $C^2(\mathbb{R}^2)$

(v) Find $f_{xy} \neq f_{yx}$ @ (0,0)

Differentiability $\Rightarrow$ all partial derivatives exist.

Find directional derivative.

Prove $f = \phi \circ g$

$$Df = D\phi \circ g + \phi \circ Dg$$

$\phi: \mathbb{R}^2 \rightarrow \mathbb{R}$
 $g: \mathbb{R}^2 \rightarrow \mathbb{R}^2$

$$f_y(x,y) = -f_x(y,x)$$

Claim: f_x & f_y are continuous at origin.

We need to know f_x in neighbourhood of origin
 & f_x at origin.

$$f_x(x,y) = \begin{cases} 0 & (x,y) = (0,0) \\ \frac{y(x^4 + 4x^2y^2 - y^4)}{(x^2+y^2)^2} & (x,y) \neq (0,0) \end{cases}$$

$$\left| \frac{y(x^4 + 4x^2y^2 - y^4)}{(x^2+y^2)^2} \right| < \frac{y((x^2+y^2)^2 + 2xy^3 - 2y^4)}{(x^2+y^2)^2} \leq \left| y \frac{3(x^2+y^2)^2}{x^2+y^2} \right| \leq (2y) \leq 3\sqrt{x^2+y^2} \leq 3\epsilon$$

$$f_y(x,y) = \begin{cases} \frac{x(y^4 + 4x^2y^2 - y^4)}{(x^2+y^2)^2} \\ 0 \end{cases}$$

$$f_{yx} = \lim_{h \rightarrow 0} \frac{f_y(h,0) - f_y(0,0)}{h} = \lim_{h \rightarrow 0} \frac{-h(4h^2 - h^4) - 0}{h} = 0$$

$$f(x,y) = \begin{cases} \frac{x^2 y}{x^2 + y^2} & \text{for } (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$$

Check for

- (i) continuity in $\mathbb{R}^2$
- (ii) Differentiability
- (iii) $g_f \in C^1(\mathbb{R}^2)$
- (iv) $g_f \in C^2(\mathbb{R}^2)$

Thm:

Let $f: \Omega \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$, Ω is open set & $a \in \Omega$

Assume that $\left\{ \begin{aligned} \nabla_{e_j} f(x) &= \frac{\partial f}{\partial x_j}(x) \end{aligned} \right\}$ exist in some ball $B(a, \delta)$

$$D_{ij} f(x) = \frac{\partial^2 f}{\partial x_i \partial x_j} = \nabla_{e_i} (\nabla_{e_j} f(x)) = f_{x_j x_i}$$

If $D_{ij} f$ exist in $B(a, \delta)$ and is continuous at "a"

then $D_{ji} f(x)$ exist and $D_{ji} f(x) = D_{ij} f(x)$

Corollary: If $f \in C^2(\Omega)$, then mixed Partial always coincide.

Observation: It is enough to prove theorem for $n=2$??

Thm: $f: \Omega \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$ $a = (x_0, y_0) \in \Omega$ (open set)

Assume that f_x, f_y and f_{xy} exist in some ball $B(a, \delta)$

Then f_{xy} is continuous at "a" then $f_{yx}(a)$ exist and $f_{xy}(a) = f_{yx}(a)$